WORKED SOLUTION

STEP II 1992, Question 14

Mechanics • Pulley

STEP II 1992, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q14

The key idea

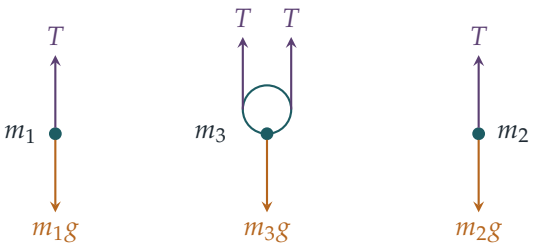
There are three moving bodies but only one string, so there is one tension T throughout and one constraint linking the three accelerations. Write Newton's second law for each particle (the pulley P_3 and its particle move together), write the string-length condition, and you have four linear equations in four unknowns. The rest condition then just says that the acceleration of P_3 is zero.

Solution

The model

The pulleys are smooth and light and the string is light and inextensible, so the tension has the same value T all along the string. Since P_3 is light, it and the particle of mass m_3 form a single body of mass m_3, held up by *two* vertical strands of string, each pulling up with force T.

Measure all displacements *downwards*: let x_1, x_2, x_3 be the depths below the fixed pulleys of the particle m_1, the particle m_2 and the pulley P_3, with accelerations a_1, a_2, a_3 (positive downwards).



Equations of motion

Resolving vertically (downwards positive) for each body:

$$\begin{aligned} m_1 a_1 &= m_1 g - T, \\ m_2 a_2 &= m_2 g - T, \\ m_3 a_3 &= m_3 g - 2T. \end{aligned}$$

The string constraint

Ignoring the fixed lengths wrapped round the pulleys, the string consists of the strand from P_1 down to m_1 (length x_1), the two strands from the fixed pulleys down to P_3 (length x_3 each) and the strand from P_2 down to m_2 (length x_2). Since the string is inextensible,

$$x_1 + x_2 + 2x_3 = \text{constant}, \qquad \text{so} \qquad a_1 + a_2 + 2a_3 = 0$$

after differentiating twice.

Ask yourself Why is the coefficient of x_3 equal to 2 and not 1?

When P_3 drops by a distance d, *both* strands supporting it get longer by d, so $2d$ of string is used up. That has to come from the outer strands, so $x_1 + x_2$ falls by $2d$. Missing this factor is the commonest way to get this question wrong.

The tension

Dividing each equation of motion by its mass,

$$a_1 = g - \frac{T}{m_1}, \qquad a_2 = g - \frac{T}{m_2}, \qquad a_3 = g - \frac{2T}{m_3}.$$

Substituting into $a_1 + a_2 + 2a_3 = 0$:

$$4g - T\left(\frac{1}{m_1} + \frac{1}{m_2} + \frac{4}{m_3}\right) = 0.$$

Multiplying through by $m_1 m_2 m_3$ and solving for T,

$$\boxed{T = \frac{4m_1 m_2 m_3\, g}{m_3(m_1 + m_2) + 4m_1 m_2}}$$

This holds throughout the motion (until a particle reaches a pulley), since nothing in the equations depends on position.

When P_3 stays at rest

The system starts from rest, so P_3 remains at rest exactly when $a_3 = 0$, that is, when $2T = m_3 g$. Using our value of T:

$$\begin{aligned} \frac{8m_1m_2m_3\,g}{m_3(m_1+m_2)+4m_1m_2} &= m_3 g \\ \iff \quad 8m_1m_2 &= m_3(m_1+m_2)+4m_1m_2 \\ \iff \quad 4m_1m_2 &= m_3(m_1+m_2), \end{aligned}$$

where we divided by $m_3 g > 0$ and multiplied by the positive denominator. So if $4m_1m_2 = m_3(m_1+m_2)$ then $a_3 = 0$, and P_3 remains at rest, as required. (Each step is reversible, so the condition is necessary as well as sufficient.)

Note that the outer particles need not be at rest: then $T = \frac{1}{2}m_3 g$ and $a_1 = g - T/m_1 = -a_2$, which is non-zero unless $m_1 = m_2$. One particle rises and the other falls at the same rate, and the string slides under P_3 without moving it.

Checks

- Symmetric case $m_1 = m_2 = m$: the condition becomes $4m^2 = 2mm_3$, i.e. $m_3 = 2m$. That is just static balance, $2T = 2mg = m_3 g$: obvious, and reassuring.
- If $m_3 \to \infty$, then $T \to \dfrac{4m_1m_2g}{m_1+m_2}$, which is twice the familiar Atwood-machine tension $\dfrac{2m_1m_2g}{m_1+m_2}$. That is right: a heavy P_3 falls freely with $a_3 \to g$, so the strands accelerate quickly. If $m_3 \to 0$ then $T \to 0$, as a massless pulley can't support any net force.
- Numerically, with $m_1 = 2$, $m_2 = 3$, $m_3 = 4$ and $g = 9.8$ (SI units), the formula gives $T \approx 21.38\,\mathrm{N}$. Solving the problem independently by an energy (Lagrangian) method gives the same accelerations $a_1 \approx -0.891$, $a_2 \approx 2.673$, $a_3 \approx -0.891\,\mathrm{m\,s^{-2}}$. With $m_1 = 1$, $m_2 = 3$, $m_3 = 3$ the rest condition holds, and indeed $a_3 = 0$ while $a_1 = -a_2 = -4.9\,\mathrm{m\,s^{-2}}$.

Common slips

- Writing the constraint as $a_1 + a_2 + a_3 = 0$, forgetting that P_3 hangs on two strands.
- Forgetting that the force on P_3 is $2T$, not T.
- Assuming that P_3 at rest forces the whole system to be at rest. It doesn't unless $m_1 = m_2$.

Beyond STEP

The constraint $x_1 + x_2 + 2x_3 = \text{constant}$ cuts the system to two degrees of freedom, and the cleanest way to handle such systems is Lagrangian mechanics: you write the energy

in terms of independent coordinates and never need the tension at all (it is a "constraint force"). This is how our numerical check worked. You'll meet it in first-year dynamics. A good place to see Newton's laws applied carefully to constrained systems such as pulleys is Kleppner and Kolenkow, *An Introduction to Mechanics*.

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